Week 10 Lab Assistant Preparation Guide

Focus: basic cybersecurity on Raspberry Pi/Linux - CIA triad, firewall basics, network traffic observation, and login/authentication logs.

Overview

Week 10 introduces cybersecurity through simple Linux tasks. Students are not doing advanced hacking. They are learning basic defensive security ideas: protecting files, checking whether data changed, keeping systems available, controlling network access, observing traffic, and checking logs for suspicious activity.

The main message for students: security is not only about passwords. It also includes permissions, integrity checks, availability, firewall rules, traffic monitoring, and log review.

Relevant repo folder and files

```
week10/
|-- 01_cia_triad.md
|-- 02_firewall.md
|-- 03_network-traffic.md
`-- 04_logins.md
```

Important: some files are labelled as Lab 5 or Lab 6 even though they are inside week10. Tell students to follow the Week 10 folder, not the title number inside the file.

1. What the week is about

- CIA triad: confidentiality, integrity, and availability.
- Using chmod to restrict file access.
- Using sha256sum to detect file changes.
- Using uptime and top to understand system availability.
- Using ufw as a beginner-friendly Linux firewall.
- Using tcpdump to observe network traffic.
- Using authentication logs and last to review login activity.

2. Relevant repo files

01_cia_triad.md

This file covers confidentiality with file permissions, integrity with hashes, and availability with system status commands.

```
nano secrets.txt
chmod 600 secrets.txt
ls -l secrets.txt
sha256sum secrets.txt
uptime
top
```

02_firewall.md

This file covers basic firewall setup using UFW.

```
sudo apt install ufw
sudo ufw allow ssh
sudo ufw enable
sudo ufw status
```

> *Teach the safer order: allow SSH before enabling UFW, especially if students are connected remotely through SSH.*

03_network-traffic.md

This file covers observing network packets with tcpdump.

```
sudo apt install tcpdump
sudo tcpdump -i any -c 20
```

The repo uses sudo tcpdump, but beginners may find that too fast. The -c 20 version captures only 20 packets and then stops automatically.

04_logins.md

This file covers authentication logs and recent login history.

```
sudo cat /var/log/auth.log
sudo grep "Failed password" /var/log/auth.log
last
```

If /var/log/auth.log does not exist, use journalctl as an alternative:

```
sudo journalctl | grep "Failed password"
sudo journalctl -u ssh
```

3. What students are expected to do

CIA triad task

- Create a file called secrets.txt and put some text inside it.
- Restrict the file with chmod 600.
- Check permissions with ls -l.
- Generate a hash using sha256sum.
- Modify the file, save it, and generate the hash again.
- Compare the two hashes and explain why they changed.
- Use uptime and top to discuss availability.
- Take screenshots and answer reflection questions.

Firewall task

- Install UFW if needed.
- Allow SSH first.
- Enable the firewall.
- Check firewall status.
- Explain why firewalls reduce unwanted network access.

Network traffic task

- Install tcpdump if needed.
- Run a short packet capture.
- Identify simple traffic details such as IP addresses, protocols, and ports.
- Explain what traffic monitoring can reveal.

Login logs task

- View authentication logs.
- Search for failed password attempts.

- Use last to see recent logins.
- Explain why logs help detect suspicious activity.

4. What you should understand before the lab

File permissions

chmod 600 means the owner can read and write the file, while the group and everyone else have no permissions.

```
chmod 600 secrets.txt
ls -l secrets.txt
```

Expected permission pattern:

```
-rw-------
```

Simple explanation: rw------- means the owner can read and write, but nobody else can access the file.

Hashing

A hash is like a digital fingerprint of a file. If the file changes even slightly, the hash should change.

```
sha256sum secrets.txt
```

Important distinction: hashing is not encryption. Encryption hides content. Hashing helps detect whether content changed.

Availability

Availability means the system is working when users need it. uptime shows how long the system has been running, and top shows active processes and resource usage.

```
uptime
top
```

To exit top, press q.

Firewall

A firewall controls which network connections are allowed or blocked. UFW means Uncomplicated Firewall and is a simpler interface for firewall rules.

```
sudo ufw allow ssh
sudo ufw enable
sudo ufw status
```

Simple explanation: a firewall is like a security guard for network traffic. It decides what is allowed in or out.

tcpdump

tcpdump shows network packets. Students may see IP addresses, protocols such as TCP, UDP, ARP or DNS, and port numbers.

```
sudo tcpdump -i any -c 20
```

> *Ethical reminder: students should only capture traffic on systems and networks where they have permission.*

Authentication logs

Authentication logs show login-related events, including successful logins, failed password attempts, SSH login attempts, sudo usage, and session openings or closings.

```
sudo grep "Failed password" /var/log/auth.log
last
```

Simple explanation: logs are like a security diary. They record what happened on the system.

5. Common student mistakes and fixes

They enable UFW before allowing SSH

If students are connected remotely, they may block their own SSH access. Teach this order:

```
sudo ufw allow ssh
sudo ufw enable
sudo ufw status
```

apt install fails

The package list may be outdated. Run:

```
sudo apt update
sudo apt install ufw
sudo apt install tcpdump
```

They do not understand chmod 600

Explain it as: owner can read and write, group has nothing, others have nothing. Then show ls -l output and point to rw-------.

The hash does not change

Usually they did not save the file, edited a different file, or did not actually change the content. Ask them to add one word, save the file, and run sha256sum again.

They confuse hashing with encryption

Correction: encryption hides content. Hashing checks whether content changed.

tcpdump output is too fast

Use a packet limit:

```
sudo tcpdump -i any -c 20
```

Or stop manually with Ctrl + C.

tcpdump gives permission denied

They need sudo:

```
sudo tcpdump -i any -c 20
```

/var/log/auth.log does not exist

Use journalctl instead:

```
sudo journalctl | grep "Failed password"
sudo journalctl -u ssh
```

grep returns no failed password results

No output can simply mean there were no matching failed password attempts. This is not automatically an error.

6. Commands, tools, and concepts to explain

```
# Edit a file
nano secrets.txt

# Save in nano
Ctrl + O, Enter, Ctrl + X

# Permissions
chmod 600 secrets.txt
ls -l secrets.txt

# Hashing
sha256sum secrets.txt
```

```
# Availability
uptime
top

# Firewall
sudo ufw allow ssh
sudo ufw enable
sudo ufw status

# Network traffic
sudo tcpdump -i any -c 20

# Logs
sudo cat /var/log/auth.log
sudo grep "Failed password" /var/log/auth.log
last
```

7. Physical or software setup to check

[ ] Raspberry Pis are powered on.

[ ] Students can access their Pi locally or through SSH.

[ ] Keyboard, mouse, monitor, or SSH access is available.

[ ] Wi-Fi or Ethernet is working.

[ ] Students know their Pi username and password.

[ ] Students know the Pi IP address if using SSH.

[ ] Internet works for apt update and package installation.

[ ] ufw and tcpdump can be installed.

[ ] /var/log/auth.log exists, or journalctl alternative is ready.

[ ] Students know how to take screenshots.

8. Simple beginner explanations

CIA triad

CIA means Confidentiality, Integrity, and Availability. It is a simple way to think about security.

Confidentiality

Only the right people can see the data. chmod 600 helps protect a private file.

Integrity

The data has not been changed without permission. sha256sum helps detect changes.

Availability

The system is working when people need it. uptime and top help check system status.

Firewall

A firewall is like a door policy. It decides which connections are allowed and which are blocked.

SSH

SSH is a secure way to remotely control the Raspberry Pi from another computer.

tcpdump

tcpdump lets us observe network traffic. It is useful for troubleshooting and security monitoring.

Logs

Logs are records of system activity. In cybersecurity, logs help us investigate suspicious behaviour.

9. Short checklist before the lab

- [ ] Week 10 repo files are available.
- [ ] Raspberry Pis are powered and accessible.
- [ ] Students can log in locally or through SSH.
- [ ] Internet works on the Pi.
- [ ] apt update works.
- [ ] ufw can be installed.
- [ ] tcpdump can be installed.
- [ ] SSH is allowed before enabling firewall.
- [ ] journalctl fallback is ready if auth.log is missing.
- [ ] You can explain chmod 600.
- [ ] You can explain hash vs encryption.
- [ ] You can explain firewall rules.
- [ ] You can explain basic tcpdump output.
- [ ] You can explain why logs matter.

10. Missing or incomplete parts in the repo

Lab numbering is inconsistent

Some files inside week10 are labelled as Lab 5 or Lab 6. Tell students to follow the Week 10 folder.

Firewall order should be safer

The safer order is to allow SSH before enabling the firewall.

```
sudo ufw allow ssh
sudo ufw enable
```

tcpdump command can be improved

Instead of sudo tcpdump, use a limited capture for beginners:

```
sudo tcpdump -i any -c 20
```

Stopping tcpdump is not clearly explained

Students should know Ctrl + C stops tcpdump.

auth.log may not exist

Prepare journalctl alternatives in advance.

No failed login attempts may appear

No grep output does not always mean a mistake. It may mean no failed password attempts are recorded.

Ethical warning is missing

Students should only capture traffic on systems and networks where they have permission.

Main teaching focus

Your main job is to help students connect commands to security concepts. Do not let them only run commands blindly. Ask: What did this command show? Why does it matter for security? Which part of CIA does it relate to?

Final summary for students: file permissions protect secrets, hashes detect changes, uptime helps us think about availability, firewalls reduce unwanted access, traffic monitoring shows what the system is doing, and logs help us investigate suspicious activity.